CartConnect: Market Basket Analysis

Detailed Implementation Guide for IBM SPSS Modeler

This document provides a comprehensive, step-by-step workflow for implementing the **CartConnect** project. Using the Apriori algorithm within IBM SPSS Modeler, we will analyze the `Groceries_dataset.csv` (comprising over 38,000 transaction records) to uncover hidden associations between products and customer purchasing behaviors.

Step 1: Project Setup and Data Import

The first objective is to load the raw transaction data into the SPSS Modeler workspace environment.

1. Launch IBM SPSS Modeler and create a new blank stream.
2. Locate the **Sources** palette at the bottom of the screen.
3. Click and drag the **Var. File** (Variable File) node onto the main canvas.
4. Double-click the node to open its properties. Under the *File* section, browse and select your `Groceries_dataset.csv` file.
5. Ensure the checkbox for **"Read field names from file"** is checked. Click *Apply*, then *OK*.

> **Pro Tip:** Right-click the Var. File node and select *Preview*. This generates a table displaying the first 10 rows, allowing you to verify that the dataset loaded correctly without delimiter issues.

Step 2: Defining Data Roles (Type Node)

Before applying the algorithm, the system must understand the specific role of each column (e.g., which column represents the receipt number, and which represents the purchased items).

1. From the **Field Ops** palette, drag a **Type** node onto the canvas.
2. Connect the Var. File node to the Type node (right-click the Source node, select *Connect*, and click the Type node).
3. Double-click the Type node and configure the fields exactly as follows:

Field Name	Measurement	Role
Transaction_ID	Categorical	**None**
Member_Number	Categorical	None
Date	Typeless / Continuous	None
Item	Categorical	**Both**

Click *Read Values* to instantiate the data, then click *OK*.

Step 3: Configuring the Apriori Algorithm

The Apriori node identifies frequent individual items in the database and extends them to larger, more complex item sets.

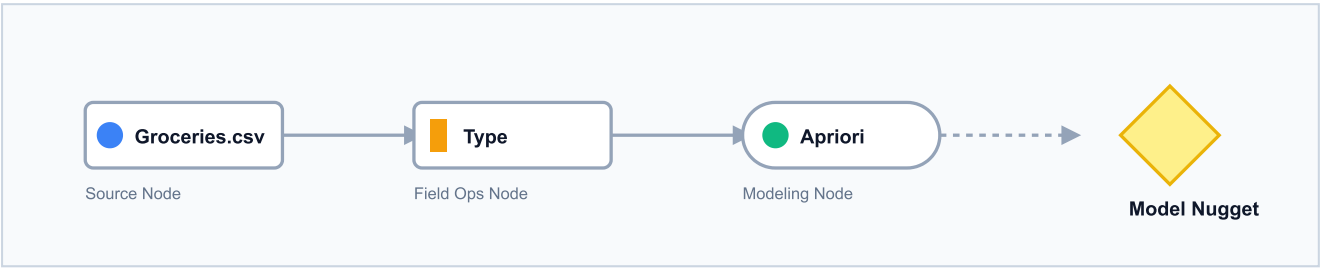
1. Navigate to the **Modeling** palette (typically located at the bottom).
2. Drag the **Apriori** node onto the canvas and connect it to your Type node.
3. Double-click the Apriori node. On the **Fields** tab, select the *"Use custom settings"* option.
4. Crucial Step: Check the box for **Transactional format**. The dataset logs one item per row, but multiple rows share the same Transaction ID.
5. Set the **ID field** to `Transaction_ID`.
6. Set the **Content field** to `Item`.

> **Parameter Tuning (Model Tab):** Switch to the *Model* tab inside the Apriori node. For grocery datasets, lower the **Minimum Support** to **1.0% or 2.0%** (since inventory is vast, few items appear in every single basket). Set the **Minimum Confidence** to **10% or 15%**.

Step 4: Executing the Stream

With the parameters configured, the stream is ready for execution. Right-click the Apriori node and select **Run**. Modeler will process the dataset and generate a Golden Model Nugget (a diamond-shaped icon) on the canvas, linked to the Apriori node.

Below is a visual representation of the final SPSS Modeler stream architecture:



Step 5: Output Analysis and Interpretation

Right-click the generated Model Nugget and select **Browse**. The system will display a list of generated association rules (e.g., `[Root Vegetables] => [Whole Milk]`). To extract business value, evaluate the rules using three primary metrics provided by the software:

Metric	Definition	Business Implication
Support	The percentage of total transactions that contain both the antecedent and the consequent items.	Indicates the overall popularity of the item combination in your store.
Confidence	Given that the antecedent was purchased, the probability that the consequent is also purchased.	Measures the reliability of the inference. High confidence suggests strong cross-sell potential.
Lift	The ratio of observed support to expected support if the items were independent.	Lift > 1 implies a positive correlation. A Lift of 2.0 means the items are twice as likely to be bought together.

Step 6: Resume Integration

When documenting the CartConnect project on your professional resume, it is vital to translate these technical SPSS steps into business-value statements. Adapt the phrasing based on the role you are applying for.

For Non-Technical / Customer Operations Roles (e.g., Teleperformance):

- "Analyzed over 38,000 transaction records to identify customer purchasing behaviors, enabling data-driven strategies for improved user experience."
- "Demonstrated strong analytical problem-solving by extracting actionable insights from complex databases using robust software tools."

For Technical / Data Science Roles:

- "Engineered a Market Basket Analysis model in IBM SPSS using the Apriori algorithm on a 38k+ row dataset."
- "Optimized model parameters (Support, Confidence, Lift) to mine transactional data for frequent itemsets and association rules."